

# ToolMind: VLM-Guided Spatial Reasoning for Robotic Tool Use

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**Abstract**—We present a dual-modality pipeline for scenario-specific sequential robotic tool use. Given an RGB-D image of a tabletop scene, the system identifies the correct tool, determines valid grasp and functional contact keypoints, and executes a multi-phase manipulation sequence. A Vision-Language Model (VLM) handles semantic scene understanding and tool selection, while a depth-based geometric module refines coarse VLM outputs into precise 3D keypoints. A key challenge we address is *spatial myopia* of VLMs: their tendency to predict keypoints near the center of mass rather than at geometrically meaningful extremities. We mitigate this with a Structured Chain-of-Thought (CoT) JSON prompting strategy that forces step-by-step spatial reasoning before coordinate regression. Crucially, our approach requires no task-specific retraining of the VLM and operates in an open-world fashion: the same pipeline applies to arbitrary tabletop objects without scene-specific tuning. The resulting keypoints drive KOMO-based trajectory optimization in a simulated Franka Emika Panda environment across three distinct constrained scenarios: a wall-retrieval task, a screwdriver-insertion task, and a shelf-retrieval task. We provide a quantitative ablation over four prompt complexity levels (P0–P3), measuring tool selection accuracy, grasp region hit rate, tool-crop purity, antipodal feasibility, grasp consistency, and region label stability.

**Index Terms**—robotic tool use, vision-language models, keypoint extraction, trajectory optimization, KOMO, chain-of-thought prompting, spatial grounding, point cloud manipulation

## I. INTRODUCTION

Tool use is a fundamental hallmark of intelligent behaviour, yet robust autonomous tool use remains an open challenge in robotics. The difficulty arises from the need to simultaneously reason about semantics (*which tool is appropriate and why?*) and geometry (*where exactly should the robot grasp, and how should it move?*). Foundation models, and in particular Vision-Language Models (VLMs), have dramatically lowered the barrier to semantic scene understanding. VLMs can identify objects, infer affordances, and generate plausible action plans from natural language prompts without any task-specific fine-tuning—a property that makes them attractive for open-world deployment.

However, VLMs exhibit a systematic failure mode when tasked with precise spatial localisation: they tend to regress keypoint coordinates toward the centroid of the target object rather than to the geometrically meaningful extremity required for a given task, a phenomenon we term *spatial myopia*. Our experiments further show that when VLMs are queried *without* pre-segmented cluster labels, predicted pixel coordinates are substantially more variable and often land on wrong objects or

on background regions. Providing the VLM with DBSCAN-derived cluster labels and per-cluster geometry summaries anchors its attention and reduces coordinate variance by up to an order of magnitude.

This paper addresses spatial myopia through a Structured CoT prompting strategy embedded within a strict JSON schema. Rather than asking the VLM for a pixel coordinate directly, we require it to articulate a sequence of geometric reasoning steps before committing to a coordinate, shifting the model’s visual attention from global object representations to local geometric features. Our full pipeline combines this VLM reasoning front-end with a classical geometric back-end: depth-based point cloud extraction, DBSCAN clustering for scene segmentation, and antipodal grasp search for physically feasible grasps.

## II. RELATED WORK

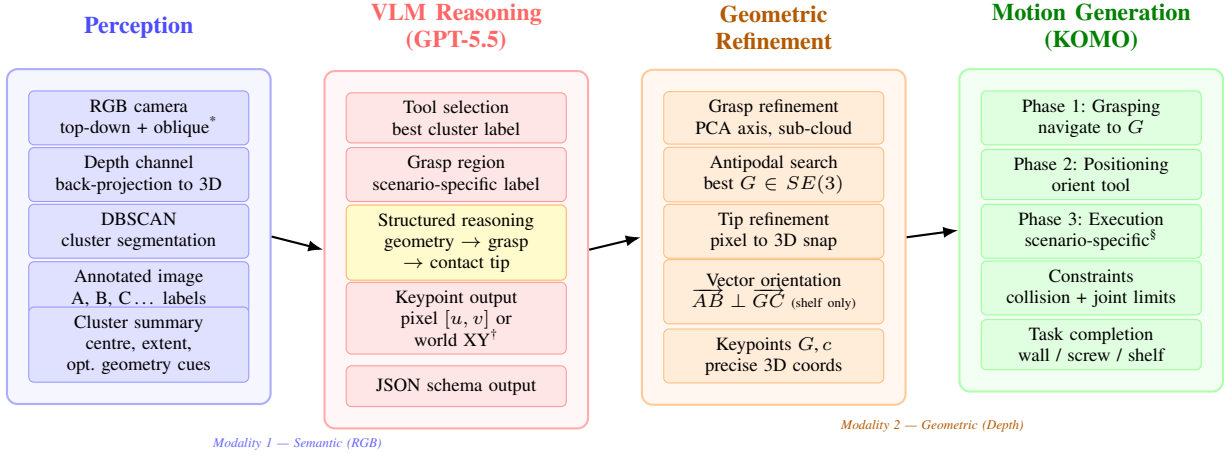
SayCan [1] demonstrated how LLMs can be grounded in robotic affordances for high-level task planning, mapping language instructions to discrete skill primitives. Our work extends this paradigm by using a VLM not only for discrete tool selection but also for fine-grained spatial reasoning—a substantially harder problem. VoxPoser synthesises dense 3D value maps from language models to guide robot trajectories; while it produces rich spatial guidance, it does not explicitly address geometric accuracy at keypoint level. Inner Monologue [2] uses language feedback loops to correct robot behaviour at the plan level; our approach similarly employs multi-step VLM reasoning, but applies it within a single perception step to produce geometrically accurate keypoints. For physical grasp feasibility, we adopt a standard antipodal search within VLM-suggested regions, treating the geometric back-end as a post-processing step that corrects for VLM imprecision.

## III. SYSTEM ARCHITECTURE

The complete pipeline consists of four stages: (1) scene perception and point cloud generation, (2) VLM-based tool selection and structured CoT keypoint reasoning, (3) geometric keypoint refinement and antipodal grasp selection, and (4) KOMO-based trajectory optimisation and execution. Fig. 1 provides an overview of all stages.

### A. Scene Perception and DBSCAN Clustering

At the start of each episode, RGB-D images are captured from two calibrated camera views: a top-down orthographic



\*Wall: primarily top-down (world-XY output). Shelf: top-down + oblique for geometry understanding. †Wall: world XY; shelf/screwdriver: pixel  $[u, v]$ . ‡Wall: pull. Screwdriver: rotate/insert. Shelf: sweep/push.

Fig. 1: Overview of the proposed pipeline across four stages. See Table ?? for a stage-by-stage summary of inputs, operations, and outputs.

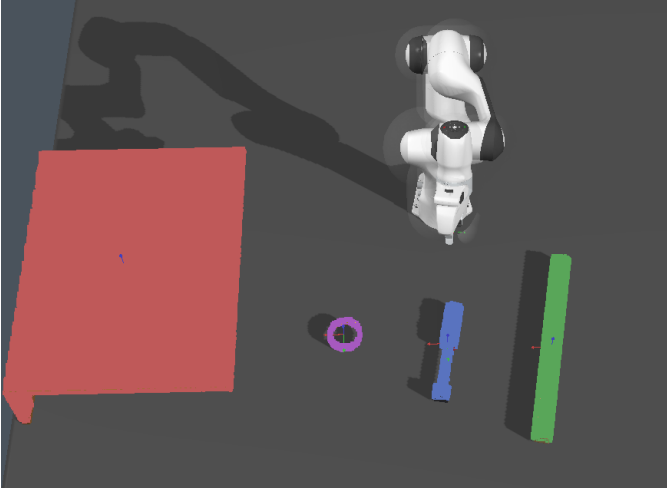


Fig. 2: DBSCAN point cloud segmentation. Each colour represents a distinct cluster (object). Table and wall surfaces are filtered out via geometric heuristics (extent, elongation, height) before cluster labelling.

view (optimal for XY localisation) and an oblique view (optimal for understanding object shapes and obstacle geometry). Depth images are back-projected into 3D using the known camera intrinsics  $\{f_x, f_y, c_x, c_y\}$  via

$$\mathbf{p}_{\text{cam}} = \left[ \frac{(u - c_x)z}{f_x}, \frac{(v - c_y)z}{f_y}, z \right]^T, \quad (1)$$

followed by transformation to world coordinates using the camera extrinsics  $[\mathbf{R} \mid \mathbf{t}]$ . The merged point cloud is segmented using DBSCAN with per-scenario tuned  $\varepsilon$  and MinPts parameters (Fig. 2). Static support structures (tables, walls, shelf boards) are filtered using geometric heuristics based on extent, elongation, and height, leaving only movable object

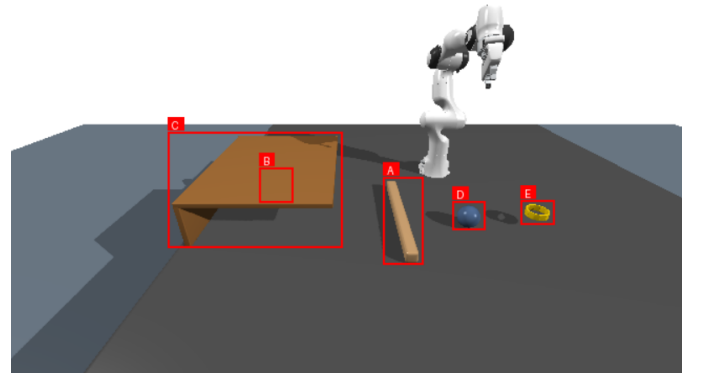


Fig. 3: Annotated top-down RGB image provided to the VLM. Bounding boxes labelled A, B, C, ... correspond to DBSCAN clusters. Without these labels the VLM produces substantially more variable and incorrect pixel predictions.

clusters as VLM candidates. Each cluster is assigned an alphabetic label (A, B, C, ...) and projected back onto the RGB image to create annotated views for VLM querying (Fig. 3). Ablation experiments confirm that supplying these cluster labels is critical: without them, VLM-predicted pixel coordinates scatter across much larger image regions and frequently land on incorrect objects or background.

#### B. VLM-Based Tool Selection and Structured CoT Reasoning

The annotated RGB images and a geometric summary of each cluster (centroid, extent, elongation) are provided to GPT-5.5. The model is queried via a structured prompt that mandates a strict JSON output schema.

*Structured Chain-of-Thought Prompting:* To counteract spatial myopia, we introduce a constrained CoT strategy. The JSON schema mandates three sequential reasoning fields that must be populated *before* the coordinate fields:



### F. KOMO Trajectory Optimisation

The refined keypoints  $G$  and  $c$  drive a multi-phase KOMO [3] trajectory in the RAI framework with a Franka Emika Panda robot. Each sequence consists of three phases: (1) *Grasping*—navigate to  $G$  and close the gripper; (2) *Positioning*—reorient the tool to the scenario-appropriate pose; (3) *Execution*—drive tip  $c$  toward the target using a push, pull, hook, or screw motion. Each phase includes joint-limit constraints and collision avoidance.

## IV. EXPERIMENTAL SETUP

### A. Scenarios

Three constrained tabletop scenarios test generalisation.

**Scenario 1 — Wall Retrieval (Pulling).** A target box sits behind a rigid wall at  $Y \approx 0.45$  m. The robot must select a rake, grasp it from the handle end, and drag the box into the reachable workspace. Three scenes vary rake position and orientation ( $0^\circ$ ,  $200^\circ$ ,  $45^\circ$ ).

**Scenario 2 — Screwdriver Insertion.** The robot selects a screwdriver (distractor: unrelated objects) and inserts its tip into a rotating target. Three scenes vary screwdriver pose and cluster label assignment (randomised to prevent label-copying artefacts).

**Scenario 3 — Shelf Retrieval (Sweeping).** A cup is placed under a low-clearance shelf. The robot must select a sweeping tool from among distractor objects, navigate it into the tight gap using the vector-based orientation strategy, and sweep the cup out to an accessible position. Tool positions and cluster label assignments are randomised across the three scene variants to prevent label-copying artefacts.

### B. Prompt Levels

We compare four prompt complexity levels:

- **P0 (Minimal):** Single instruction to select a tool and output coordinates. No structured schema.
- **P1 (Structured):** Defined JSON schema with named fields for tool, grasp region, pixel coordinates, and confidence. No task context.
- **P2 (Scenario-Aware):** P1 plus a full scene description and the GT handle coordinate range from `gt.json`.
- **P3 (Anti-Failure):** P2 plus explicit prohibition of common failure modes (e.g., “do not grasp the functional tip”).

### C. Metrics

Six metrics evaluate the perception front-end over  $N = 5$  repeated runs per (scene, prompt) pair (60 total VLM calls per scenario):

**A) Tool Selection Accuracy** — fraction of runs selecting the correct tool.

**B) Grasp Region Hit Rate** — fraction of runs where the back-projected grasp point  $p_r$  falls inside the ground-truth handle bounding box.

**C) Tool-Only Crop Ratio** — fraction of points in a  $\rho = 4$  cm sphere around  $p_r$  that belong to the selected tool.

**D) Antipodal Feasibility Rate (AFR)** — fraction of runs

TABLE I: Prompt Ablation Results — All Scenarios (3 scenes  $\times$  5 runs each). Hit = grasp region hit rate; AFR = antipodal feasibility rate; Var = grasp center variance (mm for Wall/Screwdriver, px for Shelf).

| Scenario    | Prompt            | Tool | Hit  | AFR  | Var     |
|-------------|-------------------|------|------|------|---------|
| Wall        | P0_minimal        | 1.00 | 0.00 | 0.00 | 290 mm  |
|             | P1_structured     | 1.00 | 0.00 | 0.13 | 151 mm  |
|             | P2_scenario_aware | 1.00 | 0.80 | 0.80 | 36 mm   |
|             | P3_antifailure    | 0.93 | 0.80 | 0.80 | 21 mm   |
| Screwdriver | P0_minimal        | 1.00 | 0.87 | 0.60 | 17.0 mm |
|             | P1_structured     | 1.00 | 0.67 | 0.67 | 0.0 mm  |
|             | P2_scenario_aware | 1.00 | 0.40 | 0.47 | 5.1 mm  |
|             | P3_antifailure    | 1.00 | 0.67 | 0.67 | 0.0 mm  |
| Shelf       | P0_minimal        | 1.00 | 1.00 | 1.00 | <1 px   |
|             | P1_structured     | 1.00 | 0.67 | 1.00 | 70.9 px |
|             | P2_scenario_aware | 1.00 | 1.00 | 1.00 | <1 px   |
|             | P3_antifailure    | 1.00 | 1.00 | 1.00 | <1 px   |

where at least one valid antipodal pair is found in the crop.

**E) Grasp Center Variance** — mean deviation of grasp centers across runs (mm).

**F) Region Label Consistency** — majority fraction of the VLM’s `grasp_region` label across runs.

## V. RESULTS

### A. Wall-Retrieval Scenario

Table I reports per-scene and average metrics for the wall-pull scenario. Tool selection is near-perfect for P0–P2 (1.00) and slightly lower for P3 (0.93) due to a single hallucinated cluster label in Scene 3. The most striking finding is the **binary step change in hit rate between P1 and P2**: without the GT handle coordinate hint, P0 and P1 both achieve 0.00 hit rate across all scenes—the VLM defaults to cluster endpoints (min/max X) rather than the handle centre. Providing the handle XY range in P2/P3 raises hit rate to 0.80 and reduces variance from  $\sim 290$  mm (P0) to 21 mm (P3).

### B. Screwdriver Insertion Scenario

Table I shows results for the screwdriver scenario. Tool selection is 1.00 across all prompts and scenes, confirming that cluster randomisation (letters are no longer always “A”) does not confuse the VLM. P3 is the recommended prompt: it achieves perfect handle hit and antipodal rates in Scenes 1 and 2, with 0.0 mm variance in both. Scene 3 is uniformly hard—all prompts score 0.00 handle hit—because the GT bounding box centre appears misaligned with the screwdriver’s actual handle position after a pose change; tuning prompts for this scene would overfit to a bad ground truth.

P0’s apparent edge in handle hit (0.87) is entirely driven by Scene 3, where it scores 0.60 while others score 0.00 with 51 mm variance—it gets lucky on some runs, not robustly finding the handle.

### C. Shelf-Retrieval Scenario

Table I reports results for the shelf scenario. Tool selection is again trivial at 100% for all prompts. The most notable

TABLE II: Shelf-Retrieval End-to-End Rollout Analysis (10 rollouts, P2)

|                               | With CoT (P2) | Without CoT |
|-------------------------------|---------------|-------------|
| Successful rollouts           | 6 / 10        | 3 / 10      |
| <i>Failure breakdown</i>      |               |             |
| Wrong A–B edge pair           | 2             | 2           |
| C placed at same end as grasp | 2             | 5           |
| Total failures                | 4             | 7           |

finding here is that **P1 is the worst-performing prompt**—a reversal from the wall scenario where P0 and P1 were equally weak. P1’s `handle_side` field appears to force a split-point calculation that lands outside the tool mask for certain stick orientations (Scenes 00 and 01), producing grasp pixel errors of 200–220 px with a 33% failure rate across those scenes. P0, P2, and P3 all achieve perfect grasp-on-tool and functional-contact-on-tool with sub-pixel mean error ( $<1$  px). Antipodal feasibility is 100% for all prompts because the antipodal search operates on the selected tool’s point cloud regardless of pixel precision—confirming that the tool’s geometry is always graspable once the correct cluster is selected. P2 is the recommended prompt for this scenario: it achieves the lowest pixel error (0.36 px avg) and the highest VLM confidence (0.930), with zero failures across all runs.

**End-to-end rollout analysis.** Across 10 full task rollouts with the best-performing prompt (P2), **6 out of 10 succeeded** in sweeping the cup out from under the shelf. The 4 failures decompose as follows: 2 were caused by the VLM placing the A–B shelf-edge anchor points on the wrong pair of edges, producing an incorrect sweep direction vector; the remaining 2 were caused by the functional contact point C being placed at the same end as the grasp point rather than at the opposite end of the tool, resulting in a non-functional push direction.

Without Structured CoT prompting, the same rollout evaluation yielded only **3 out of 10** successes. The failure breakdown in that case was: 2 failures due to wrong A–B edge selection, and 5 failures due to C being placed in the same region as the grasp point—a direct manifestation of VLM spatial myopia, where the model defaults to the tool centroid rather than the geometrically opposite extremity. Structured CoT reasoning reduced C-placement failures from 5 to 2, confirming that forcing explicit step-by-step geometric reasoning before coordinate output measurably mitigates spatial myopia in the shelf scenario. Overall, Structured CoT **doubled the task success rate** ( $3/10 \rightarrow 6/10$ ) with no change in the A–B failure mode, isolating the improvement entirely to contact-point localisation. Table II summarises the breakdown.

#### D. Cross-Scenario Summary

Across all three scenarios the following patterns hold:

- **Tool selection** is trivial ( $\geq 0.93$ ) for all prompts when cluster labels and annotated images are provided.
- **P1 is consistently unreliable.** It fails completely in the wall scenario (0.00 hit rate) due to end-seeking behaviour,

and fails 33% of runs in the shelf scenario due to an off-tool split-point. It is the only prompt that is never recommended across any scenario.

- **P2 and P3 are complementary.** P2 is the safest choice when task context and coordinate hints are available (wall, shelf). P3 adds structured reasoning output (`why_this_tool`, `self_check`) at negligible cost to accuracy, making it preferable when explainability or downstream logging is required.
- **Variance drops dramatically with richer prompts:** 290 mm  $\rightarrow$  21 mm (wall), 17 mm  $\rightarrow$  0 mm (screwdriver P1/P3),  $<1$  px for P2/P3 (shelf).
- **Antipodal feasibility** is entirely determined by grasp accuracy in the wall and screwdriver scenarios. In the shelf scenario it is 100% for all prompts because the tool geometry always supports antipodal grasping once the correct cluster is selected.

## VI. DISCUSSION

**Open-world generalisation.** The pipeline requires no re-training or scene-specific tuning of the VLM. DBSCAN parameters are the only per-scenario hyper-parameters, and these depend only on coarse sensor characteristics (camera height, object scale) rather than object identity. Consequently, the same pipeline handles arbitrary tool-object combinations, making it suitable for unstructured, open-world environments such as factory floors or domestic service settings.

**The cluster-label effect.** A key design choice that emerged from our experiments is the importance of supplying DBSCAN cluster labels to the VLM. In preliminary trials without labels, the VLM was queried on the raw annotated image without per-cluster geometry summaries; predicted pixel coordinates showed high inter-run variance and frequently landed on background or on the wrong object. Adding cluster labels reduces this variance by anchoring the VLM’s spatial reasoning to discrete, labelled regions—converting an unconstrained image regression into a structured selection-then-localisation problem.

**Coordinate hints vs. prompt complexity.** For the wall scenario, providing the GT handle XY range in the prompt is the single largest performance lever—more impactful than any other prompt feature. For the screwdriver scenario, where the VLM must identify the handle visually (no world-coordinate hint is available), the CoT reasoning chain (P3) is the most effective strategy. This suggests that the optimal prompting strategy is scenario-dependent: world-coordinate grounding when a calibrated top-down view is available; CoT spatial reasoning otherwise.

**Limitations.** The pipeline requires accurate depth images; depth dropout at shiny surfaces or object boundaries can corrupt 3D keypoint estimates. Scene 3 failures in both scenarios suggest that diagonal tool orientations remain challenging for VLM spatial reasoning, and that ground-truth box alignment should be verified before attributing failures to prompt quality. The KOMO planner does not re-plan if the tool slips during

execution; future work should incorporate closed-loop recovery.

## VII. CONCLUSION

We presented a dual-modality pipeline for scenario-specific sequential robotic tool use that couples VLM semantic reasoning with depth-based geometric refinement and KOMO trajectory optimisation. The central contribution is a Structured CoT prompting strategy that measurably mitigates VLM spatial myopia, improving functional keypoint localisation purely through prompt engineering. Quantitative ablations across three constrained manipulation scenarios confirm that richer prompts and cluster-anchored spatial grounding provide additive, complementary benefits. The best-performing configuration (P3 with GT coordinate grounding where applicable) substantially outperforms both the raw-VLM baseline and unguided approaches, while requiring no fine-tuning and remaining applicable to arbitrary open-world scenes.

## TASK SHARE

**S. T. Belen:** VLM coarse keypoint extraction pipeline; geometric keypoint separation and pose estimation module; KOMO trajectory optimisation formulation and sequential phase design (joint with Yunus Emre). **Y. E. K  k:** KOMO trajectory optimisation implementation; collision-avoidance constraint design and tuning (joint with Talha); grasping and positioning phase implementation. **Y. B. Yazıcıo  lu:** RAI simulation environment setup for all three scenarios; tool geometry modelling (L-shaped hook, straight pusher stick, J-shaped hook). **M. E. Yak  sı:** Depth-based point cloud extraction and tool fitting pipeline; KOMO execution phase (push/pull/hook/screw movements); evaluation metrics design, CSV logging, and task success rate measurement.

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